

One Size Does Not Fit All: Methodological Considerations for Brain-Based Predictive Modeling in Psychiatry

Elvisha Dhamala, B.T. Thomas Yeo, and Avram J. Holmes

ABSTRACT

Psychiatric illnesses are heterogeneous in nature. No illness manifests in the same way across individuals, and no two patients with a shared diagnosis exhibit identical symptom profiles. Over the last several decades, group-level analyses of in vivo neuroimaging data have led to fundamental advances in our understanding of the neurobiology of psychiatric illnesses. More recently, access to computational resources and large, publicly available datasets alongside the rise of predictive modeling and precision medicine approaches have facilitated the study of psychiatric illnesses at an individual level. Data-driven machine learning analyses can be applied to identify disease-relevant biological subtypes, predict individual symptom profiles, and recommend personalized therapeutic interventions. However, when developing these predictive models, methodological choices must be carefully considered to ensure accurate, robust, and interpretable results. Choices pertaining to algorithms, neuroimaging modalities and states, data transformation, phenotypes, parcellations, sample sizes, and populations we are specifically studying can influence model performance. Here, we review applications of neuroimaging-based machine learning models to study psychiatric illnesses and discuss the effects of different methodological choices on model performance. An understanding of these effects is crucial for the proper implementation of predictive models in psychiatry and will facilitate more accurate diagnoses, prognoses, and therapeutics.

<https://doi.org/10.1016/j.biopsych.2022.09.024>

Progress across the brain sciences has provided unprecedented opportunities to advance our understanding of the etiology and pathogenesis of psychiatric illness. In the past several decades, new in vivo imaging techniques, such as magnetic resonance imaging (MRI), have facilitated the successful identification of neural correlates of human behavior. Traditionally, patient populations were treated as distinct entities, and corresponding group-level analyses shed light on the neurobiological changes that characterize these nominally distinct diagnostic constructs. However, the heterogeneity in psychiatric illnesses and murky boundaries between health and disease make it difficult to establish the extent to which brain-behavior relationships are consistently or uniquely disrupted. To address the heterogeneous nature of psychiatric illnesses and circumvent diagnostic biases, precision medicine has emerged as a powerful tool to understand clinical presentation at an individual level by considering a range of biological and environmental factors (1). We are currently at an exciting time when we can harness the joint powers of precision medicine, machine learning, and neuroimaging to enhance our understanding of the neurobiology of psychiatric illnesses and associated symptom profiles. Increased access to computational resources and large, publicly available datasets are supporting a field-wide shift to data-driven discovery science, allowing descriptive models of psychiatric illness, predictions of clinically relevant behaviors, and

personalized prescriptions for therapeutic interventions ushering in a new era for psychiatry.

In psychiatry, precision medicine and predictive modeling have 2 primary goals: to discover how the brain gives rise to behavior and to identify practically useful markers of behavior and clinical outcomes. Precision medicine transitions away from the one-size-fits-all model of traditional medicine and seeks to identify the underlying causes of disease and appropriate therapeutic interventions at a patient-specific level. This individualized approach has led to significant advances across the health sciences, for instance in oncology, where personalized treatments are now available for a range of cancers (2). The application of predictive models in psychiatry, termed precision psychiatry (3), incorporates insights from molecular biology, cognitive neuroscience, neurophysiology, and behavioral and clinical psychology to capture clinically relevant biomarkers. If successful, precision psychiatry will transform how we diagnose and treat psychiatric illnesses.

There is growing interest in the development and application of novel tools to predict illness risk and disease progression and to identify shared and unique properties of diagnostic groups and associated symptom profiles. Psychiatric diagnoses categorize clusters of co-occurring symptoms that often lack clear, discernible borders between health and disease (4). While the presence of specific symptom profiles can distinguish diagnoses, phenotypic presentations can overlap

across patient groups, making it difficult to identify specific markers of illness. In this regard, discrete and dimensional models of psychiatric illness provide complementary information. Discrete medical models provide diagnostic labels based on clinical descriptions of signs and symptoms, while dimensional models describe broad hierarchical features of psychopathology. Together, different models of psychiatric illness can facilitate improved understanding of psychopathology at an individual level.

Traditional analyses in psychiatry have largely sought to uncover group-level patterns characterizing distinct illnesses. While these analyses have provided foundational insights in our understanding of psychiatric illness, they fail to adequately capture individual variability in symptom presentations. Machine learning provides a rich analytic framework to reveal the neurobiological correlates of discrete and dimensional measures of psychopathology at an individual level. By leveraging our understanding of brain-behavior relationships and considering a specific patient's personalized profile, machine learning can be used to generate individual-level behavioral predictions. Machine learning analyses can be characterized into 3 groups—descriptive, predictive, and prescriptive—each with distinct utility for psychiatry researchers (Figure 1). These analyses rely on supervised, semisupervised, and unsupervised algorithms. Because an exhaustive overview of different analyses and algorithms is not possible here, we provide key terms and examples in Table 1 [for additional discussion, see (5)].

Guidelines for implementing predictive models to study behavioral variability based on neuroimaging data have been

proposed (5–11), but there has not yet been a review of how different methodological considerations can influence the models. Here, we cover the effects of algorithm selection, neuroimaging modality, data transformation, sample sizes, brain parcellation, phenotype selection, and population-specific approaches on predictive models and contextualize how insights gained from those analyses can facilitate advances in psychiatry (Figure 2).

METHODOLOGICAL CONSIDERATIONS

Algorithm Selection

The structure of neuroimaging data presents a unique set of analytic challenges in terms of dimensionality, collinearity, and sparsity. To address these challenges, different algorithms have been designed including the connectome-based predictive model (12) and BrainNet convolutional neural network (13). The benefits of these cutting-edge approaches include 1) data-driven feature selection algorithms that do not rely on brain regions or connections selected a priori, but rather account for the fact that behavior and cognition may be best characterized as properties of covariation across large-scale networks; 2) built-in cross-validation, which ensures that models are generalizable and replicable; and 3) the potential for continuous predictions that, in contrast to categorical or diagnostic predictions, provide more dimensional measures of psychopathology. Connectome-based predictive modeling is a data-driven protocol to predict behavior from connectivity data. Here, researchers determine which connectivity features are most strongly correlated with a given phenotype and then use those features to build a linear

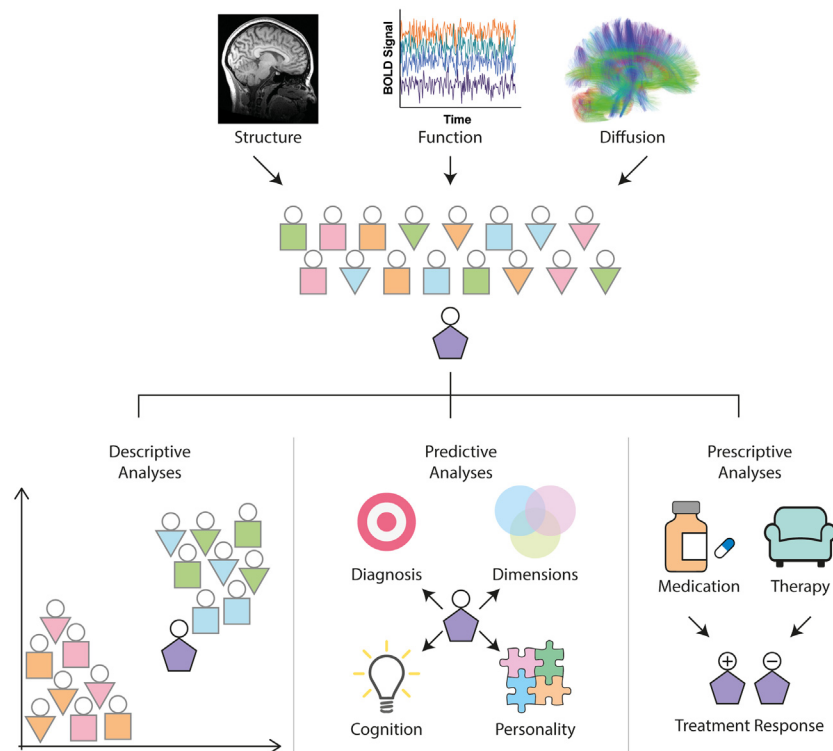


Figure 1. Predictive models based on neuroimaging data can be applied in psychiatry to perform descriptive, predictive, and prescriptive analyses. Information obtained from structural, functional, and diffusion scans at a population level can be leveraged to gain insights at an individual level. Descriptive analyses detect underlying patterns in the neurobiological data at a population level and describe how those patterns manifest at an individual level. These approaches rely largely on unsupervised algorithms and are technically postdictive in that they retrospectively detect trends in existing data rather than prospectively make predictions. Predictive analyses capture brain-behavior relationships at a population level and use those to predict psychiatric diagnosis, dimensions, cognition, and personality at an individual level. These predictive models rely on supervised algorithms and have the potential to identify future risks for development or deterioration of illness in a personalized manner. Prescriptive analyses establish neurobiological predictors of treatment response trends at a population level and provide recommendations on interventions at an individual level. These approaches rely on supervised algorithms to identify anomalies and suggest personalized interventions that can most effectively alleviate individual-specific patterns of concerns and/or other relevant factors. Together, descriptive, predictive, and prescriptive analyses can yield clinically meaningful insights that can facilitate the development of personalized diagnostic tools and therapeutic interventions. BOLD, blood oxygen level-dependent.

Table 1. Definitions of Key Terms

Category	Term	Definition/Description	Examples and Additional Resources
Analyses	Descriptive	Rely on existing observations to identify underlying patterns that can describe and summarize the data in an interpretable manner	Identification of distinct subtypes of autism (132,133), depression (134,135), psychosis (136), schizophrenia (33,137), and bipolar and schizoaffective disorders (138)
	Predictive	Establish associations between independent and dependent variables in existing data, use learned associations to predict trends and properties in unobserved samples	Prediction of clinical, cognitive, and personality traits across different psychiatric illnesses (30,139–142)
	Prescriptive	Identify patterns and trends in the underlying data and recommend personalized interventions that can most effectively alleviate individual-specific patterns of concern	Prediction of clinical responses to medications (18,41) and cognitive behavioral therapy (36,46,47)
Algorithms	Supervised	Rely on preobserved ground truth labels to learn associations between the input and output data	Prediction of treatment response in social anxiety disorder (46)
	Semisupervised	Combine a small amount of labeled data and a large amount of unlabeled data to identify associations between the input and output data	Identification of subtypes of internalizing disorders in youth (143)
	Unsupervised	Group and interpret patterns in unlabeled data to characterize data	Investigation of overlapping and specific correlates of common psychiatric illnesses (144)
Supervised Algorithms	Classification	Establish relationships between the input variables and discrete binary or multiclass categorical labels	Distinguishing healthy and clinical populations; see (145) for a review on the application of clinically informative classifiers
	Regression	Predict continuous variables based on the input data	Prediction of disease severity; see (146) for a review on the use of regression algorithms to elucidate brain-behavior relationships
Unsupervised Algorithms	Clustering	Enable the characterization of data into distinct groups (clusters) based on shared underlying patterns	Biologically driven disease subtype identification
Models	Classical machine learning	Relatively simple models that rely on linear or nonlinear and sparse or nonsparse approaches to derive associations between the input and output data	Linear regression, logistic regression, support vector machines
	Deep learning	Subset of machine learning models that rely on one or more neural networks (i.e., perceptrons) to progressively extract higher-level features from the input data to make predictions about the output	Feedforward neural networks, convolutional neural networks
Variables	Binary	Discrete data that can fall in one of 2 categories	Presence or absence of an illness
	Categorical	Discrete data that can fall in one of several categories	Subtypes of an illness
	Continuous	Data can take on any value within a continuous range	Dimensional measures of impairment
Regularization	L1 (sparse)	Uses the absolute value of magnitude of the parameters as the penalty term, thus encouraging sparsity by shrinking least important coefficients to 0	
	L2 (nonsparse)	Uses the squared magnitude of the parameters as the penalty term, thus reducing the magnitudes of parameters	
Model Properties	Complexity	Capacity to capture more complicated relationships among the features and target variables	
	Transparency	Ability to understand and interpret the brain-behavior relationships captured by the model	
Model Performance	Accuracy	Correlation (or correspondence) between true and predicted values when evaluating a model on a (held-out) test set that is from the same population (or dataset) as the training set	
	Generalizability	Correlation (or correspondence) between true and predicted values when evaluating a model on a population (or dataset) that is different from the population in which it was trained	

model to predict behavior. This approach has identified functional connectivity correlates of attention (14,15), memory (16), and anxiety (17) across distinct populations and predicted responses to antidepressants (18) and treatment for cocaine and opioid use disorders (19,20). BrainNet relies on a convolutional neural network framework that maintains the topological complexity of brain networks. This framework has been applied

to predict future cognitive and motor development in preterm infants based on white matter connectivity (13).

While both connectome-based predictive models and BrainNet convolutional networks support the analysis of connectomic data, other classical machine learning and deep learning algorithms can also generate meaningful predictions. Amid ongoing debates about which algorithms yield the most

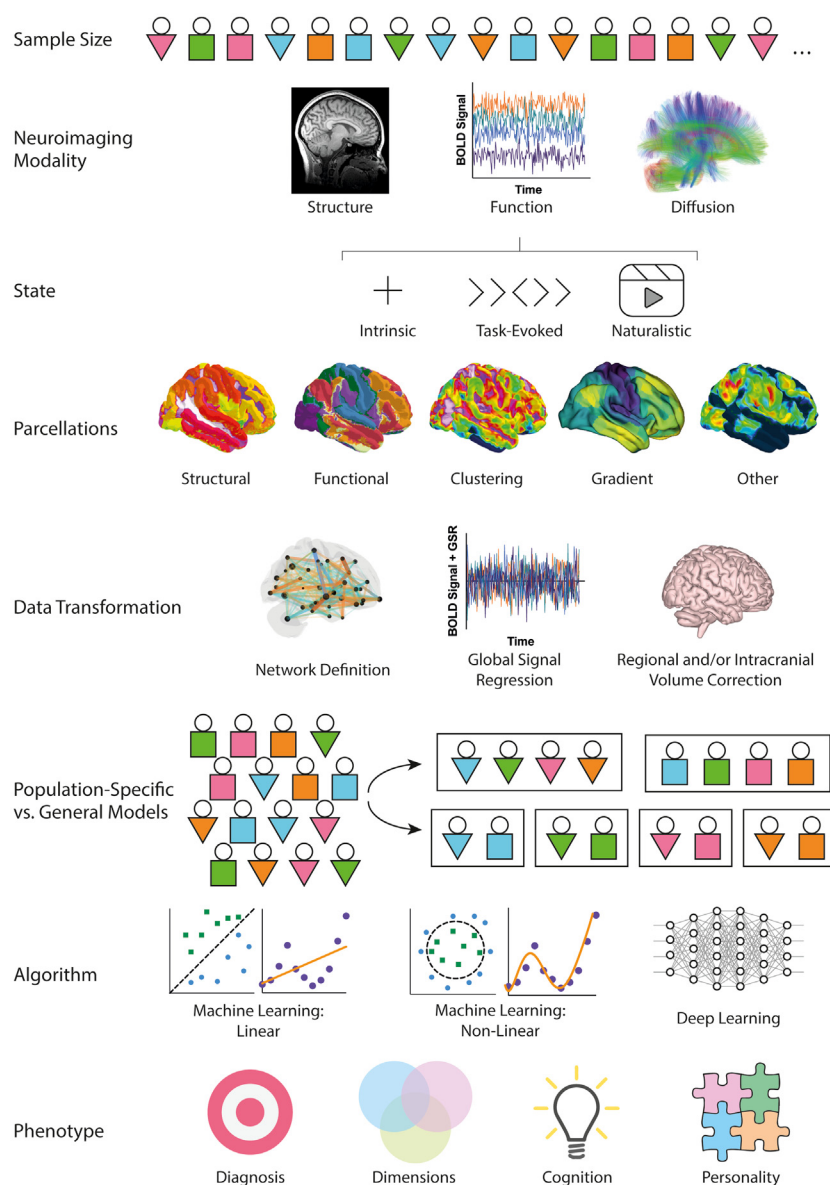


Figure 2. Methodological choices for machine learning models based on neuroimaging can bias the performance of predictive models. Sample size: the number of individuals and the heterogeneity within the population. Neuroimaging modality: the choice of structural, functional, or diffusion imaging scans. State: functional scans may be acquired to measure activation patterns corresponding to intrinsic (resting-state), task-evoked, or naturalistic (e.g., movie watching) states. Parcellations: distinct parcellations derived using structural, functional, clustering, gradient, or other information may be applied to the neuroimaging data. Data transformation: neuroimaging data must be preprocessed and transformed before it can be input into a predictive algorithm. Population-specific vs. general models: the entire sample may be considered as a single population or separated into distinct subsamples based on different factors such as demographics or diagnosis. Algorithm: predictive algorithms include linear or nonlinear traditional machine learning and deep learning approaches. Phenotype: the specific variable to be predicted may be a categorical diagnosis, dimensional measures of psychopathology, cognitive abilities, or personality traits. Predictive modeling in psychiatry must be designed and implemented with careful consideration of how these factors may potentially influence model accuracy and interpretability. BOLD, blood oxygen level-dependent. GSR, global signal regression.

accurate predictions, head-to-head comparisons suggest that a suite of approaches may be clinically useful. In one recent comparison, kernel ridge regression, a generic feedforward network, a BrainNet convolutional network, and a graph convolutional network generated comparable cognitive predictions based on functional connectivity (21). A separate analysis also reported that classical approaches, including elastic net (which optimizes between L1 and L2 regularization), achieve comparable or better predictive accuracies than deep learning methods to capture associations between functional connectivity and cognition (22). However, findings have been mixed, and a separate comparison revealed that deep learning models produced the best cognitive predictions based on

whole-brain gray matter volume in healthy and clinical (mild cognitive impairment, Alzheimer's disease) populations (23).

Comparisons of classical and deep learning approaches for classification problems have also produced mixed findings. One study found that deep learning classifiers were more accurate when relying on anatomical or functional data (or both) to distinguish individuals with autism from healthy control subjects (24). On the contrary, an international challenge to predict autism spectrum diagnosis from anatomical and functional data revealed that deep learning models were more likely to overfit such that they failed to generalize to different datasets, whereas simpler models achieved more robust performance (25). Relatedly, across different classical

algorithms, nonsparse linear methods more accurately distinguish different psychiatric diagnostic groups from healthy control subjects than sparse linear or nonlinear methods (26).

Therefore, different machine learning algorithms are suitable for distinct predictions, and algorithm selection should consider whether accuracy, interpretability, or both should be prioritized. Critically, these decisions should be primarily driven by the specific study goals.

Neuroimaging Modality

Different neuroimaging modalities capture distinct information about the underlying neurobiology. Structural MRI provides information about the shape, size, and integrity of gray and white matter brain structures and can be used to study physical damage and lesions. Functional MRI indirectly measures brain activation by detecting changes in blood oxygenation, facilitating the study of functional activation patterns and correlated signals across functional systems. Finally, diffusion MRI measures diffusion of water molecules in biological tissues and enables the quantification of white matter tracts throughout the brain. These modalities yield distinct biologically relevant features for the study of psychiatric illnesses and prediction of clinically relevant phenotypes.

Converging evidence suggests that individual differences in cognition, personality, and emotional traits, as well as psychiatric diagnoses, are most accurately predicted by functional connectivity (24,25,27–31). These results, reported across healthy and clinical populations, suggest that functional MRI data may be the best modality for predictive modeling. However, there is evidence to suggest that other modalities may also produce clinically informative predictions, especially in psychiatric illnesses associated with structural abnormalities in the brain.

Cortical gray matter volume and thickness successfully predict diagnosis of Alzheimer's disease, autism, and attention-deficit/hyperactivity disorder (32) and symptom burden in schizophrenia (33). Gray matter density reliably estimates response to repetitive transcranial magnetic stimulation in patients with schizophrenia (34). Finally, voxel-level structural scans identify whether a patient with depression may be treatment resistant (35).

Meanwhile, diffusion-derived measures of fractional anisotropy along with radial, mean, and axial diffusivity predict response to cognitive behavioral therapy in social anxiety disorder (36). White matter connectivity characterizes tic improvement following deep brain stimulation for Tourette syndrome (37) and distinguishes healthy control subjects from patients with mild cognitive impairments and patients with dementia (38).

Intrinsic Versus Task-Evoked Measures of Brain Functioning

Intrinsic (or resting-state) functional connectivity captures the extent to which neural activation patterns between a pair of brain regions covary over time, providing insight into general functional network organization in the brain. Task-evoked functional connectivity characterizes covariations in neural activation patterns while performing specific tasks and

facilitates understanding of the functional networks that underlie different behaviors. While both intrinsic and task-evoked measures of functional network connectivity can generate insightful behavioral predictions, models based on task-evoked connectivity predict individual cognitive traits more successfully (39,40). Moreover, stacked models incorporating multimodal data yield the greatest accuracy (29). Intrinsic striatal connectivity predicts responses to antipsychotic drug treatment in patients with schizophrenia and acute psychosis (41), while whole-brain intrinsic functional connectivity distinguishes bipolar and major depressive disorder (42) and estimates risk of developing anxiety disorders (17). Meanwhile, connections evoked during emotional face perception tasks meaningfully distinguish individuals with social anxiety disorder and individuals with panic disorder from healthy control subjects (43). Task-evoked connectivity using memory and general executive function tasks also transdiagnostically predicts memory deficits across schizophrenia, bipolar disorder, and attention-deficit/hyperactivity disorder, and these models generalize to healthy participants (16). Alternatively, task-evoked measures of brain activation are also clinically meaningful. Activation patterns discriminate between individuals with schizophrenia and bipolar disorder (44), distinguish major depression with and without mania from bipolar depression (45), and predict clinical responses to cognitive behavioral therapy in social anxiety disorder (46) and panic disorder (47).

The acquisition of high-quality functional MRI scans sensitive to motion may be difficult to achieve in psychiatric populations. Patient well-being can be prioritized while addressing this issue, and high-quality data can be acquired using different strategies including familiarizing subjects with the scanner environment, using physical cushioning/restraints, and implementing motion-correction strategies (48–51). Moreover, while task scans predict behavioral traits more accurately in healthy populations, they can be more cognitively demanding on participants, require additional scan time, and present language barriers, making them less feasible in clinical samples. To circumvent this issue, resting-state data can be used to synthesize task data (52) and naturalistic stimuli (movie watching) data (53). Those synthesized data can predict individual differences in brain activity during task performance (52), naturalistic viewing (53), and behavioral traits (54) in healthy populations. In a clinical context, this approach can facilitate more accurate behavioral predictions in a psychiatric population, reducing (and even eliminating) the monetary and cognitive resources associated with the acquisition of task and naturalistic scans.

Data Transformation

To transform noisy estimates of brain structure and function into reliable predictors of human behavior, neuroimaging data must be carefully preprocessed and cleaned. Preprocessing pipelines and analytical methods can introduce variability into neuroimaging studies and lead to biased findings (55–58).

Before any processing, analyses, or modeling can be performed, neuroimaging data must undergo quality control to ensure that images are free of artifacts and have a high signal-to-noise ratio. Unfortunately, stringent quality control is likely

to result in samples that are less representative of the population in terms of age, sex, race, and socioeconomic status and remove meaningful variance in the data (40). Lower data quality may also be indicative of and associated with psychiatric illness. Hence, careful consideration must be given to arbitrary quantitative and qualitative data quality thresholds that may bias subsequent predictive modeling analyses.

Relatedly, missing imaging and behavioral data are also a particular point of concern in clinical populations. Missing data may be a direct result of data entry errors, participant difficulty in completing data collection, attrition at follow-up, or an unintended consequence of stringent quality control (59). Similar to quality control, missing data are more likely to be present in clinical populations, especially those with severe psychiatric illness, and individuals from specific demographic groups who are likely already underrepresented. Although potential solutions for missing data imputation have been proposed, the field has not yet reached a consensus about best practices. Future research should incorporate lessons learned from decades of clinically oriented research seeking to capture complex clinical outcomes using incomplete data.

Global signal regression is a highly debated method used to remove the effects of global variations in activity from the functional time series of each voxel (60,61). Most, although not all, behavioral predictions across cognitive, personality, and emotional domains based on intrinsic and task-based functional connectivity benefit from global signal regression (39,62).

Another critical analytical choice for functional MRI addresses the parametrizations of the functional connectome. Full correlation is a standardized form of covariance. Partial correlation is derived using the inverse of the precision matrix and represents the correlation between a pair of brain parcels after adjusting for the time series of all other parcels. Furthermore, not every matrix is a valid correlation (or covariance) matrix. Tangent space representations allow manipulation of the functional connectivity data consistent with the mathematical properties of correlation (or covariance) matrices (22). Some studies have suggested that compared with full and partial correlations, tangent space representations yield more accurate and reliable diagnostic classifications across distinct psychiatric illnesses (26).

Relatedly, white matter connectivity is defined using deterministic and probabilistic tractography algorithms (63,64). Deterministic tractography relies on fixed orientations to direct streamlines, while probabilistic algorithms estimate a distribution of fiber orientations and randomly draw a sample from this distribution to direct streamline propagation (64). White matter connectivity derived using probabilistic tractography predicts cognition more accurately and reliably than deterministic tractography (27). Taken together, these results suggest that connectomic definitions can also bias behavioral predictions.

Additionally, the use of anatomical features for predictive modeling involves crucial data transformation steps that can bias behavioral predictions. Individual differences in intracranial volume are typically considered to be confounds (65), but proportional correction of those individual variations can differentially bias cognitive predictions based on cortical surface area, gray matter volume, and thickness (66). To better

capture clinical deviations in brain structure and function, normative modeling can be used to model expected variations in a given neuroimaging feature (67). Deviations from normative models of regional cortical volumes more accurately predict overall and dimension-specific psychopathology than raw measures of cortical volume (68).

Critically, global brain activation (69), intracranial volume (70,71), and regional cortical volume (72–75) have been implicated in and may be indicative of psychiatric illness. Therefore, it is crucial to account for the underlying, clinically relevant signals that may be captured in distinct properties and consider how they should be handled when developing predictive models to study psychiatric populations.

Phenotype Selection

An unresolved debate pertains to the structure of psychiatric illnesses and whether researchers should focus on diagnoses, symptoms, or dimensional profiles (76–78). Psychiatric diagnoses rely on a clinician's description of the syndrome's nature, time span, severity, and preexisting risk factors to establish a categorical classification (79). These descriptions may be biased, and the resulting diagnoses unreliable. In fact, two of the most common psychiatric illnesses, major depressive disorder and generalized anxiety disorder, have interrater reliabilities of 0.28 and 0.21, respectively (80). Therefore, clinically distinct phenotypic profiles can be assigned the same diagnosis. Remarkably, there are 636,120 distinct clinical presentations of posttraumatic stress disorder (81). It is naïve to believe that the same neurobiological properties will underlie the unique clinical presentations of a given diagnosis (82). Further affecting the clinical utility of brain-behavior models, clinical scales can vary substantially in terms of the symptoms and dimensions they capture, and single symptomatic or dimensional measures are insufficient to capture the complex nature and presentation of a psychiatric illness in an individual. These data raise many questions about which phenotypes or diagnoses we should aim to predict and whether those predictions will be biologically meaningful. Data scientists are familiar with the idea of “garbage in, garbage out”: If relying on poor phenotypic data and/or labels to train predictive models, the predictions and underlying associations captured are also likely to be of poor quality. Therefore, to maximize the accuracy and interpretability of prediction models in psychiatry, the variables being predicted must be reliable, validated, and representative of the overarching psychiatric illness or phenotype being studied in the specific population of interest.

Directly addressing this pitfall, researchers have begun to develop frameworks to reconceptualize diagnostic criteria for psychiatric illnesses into quantifiable behavioral measures, such as the Extended Strengths and Weaknesses Assessment of Normal Behavior (83). Moreover, the use of passive digital data collection (i.e., from smartphones and wearable technology) along with better-distributed self-report measures is likely to yield more representative and robust quantifications of clinical behaviors. Overall, an improvement in the quantification of complex behaviors is crucial for the successful implementation of predictive models to capture relationships between those behaviors and neurobiological markers.

Ongoing research also suggests that certain illnesses, behaviors, or phenotypes may be more predictable than others. Intrinsic and task-based functional connectivity and structural properties yield higher accuracies when predicting cognitive traits than personality or mental health traits (40,84). Within the cognitive domain, composite scores summarizing broad cognitive domains (85–87) are predicted more accurately than individual task scores (88). Therefore, in addition to a variable being valid, reliable, and of clinical interest, its associations with neurobiological properties (and the underlying signal-to-noise ratio) must be strong enough to be captured using multivariate approaches.

Relatedly, evaluation of model generalizability across datasets can be tricky because they may collect different clinical scales and psychometric measures. Fortunately, recent attempts to characterize patient profiles using symptomatology and behavioral measures have revealed that specific connectivity patterns underlie multiple phenotypic expressions (89). While further research is needed in this area, these findings suggest that in some instances, a single model may be generalizable across datasets or populations in predicting different independent, but phenotypically correlated, measures.

Brain Parcellations

Machine learning analyses using neuroimaging data are prone to the curse of dimensionality. Imaging data at a voxel level contains hundreds of thousands of features. Brain parcellations reduce dimensionality by segmenting the brain into tens or hundreds of regions or networks (90,91). These parcellations may be defined based on cytoarchitectural organization (92–94), sulcogyral localization (95–97), data-driven modeling of intrinsic (98–103) or task-evoked (104) functional activation signals, or independent component analyses (105,106). Dimensionality reduction through parcellation reduces noise, decreases computational complexity, and enhances interpretability (107), but it can also result in loss of biologically meaningful signals captured at a voxel level and obscure valuable individual-level variations in brain organization (107–109). While some analyses have shown evidence that parcellation resolution and definition can affect predictive performance (26,27,110,111), others have reported generally consistent results across parcellations (24).

To eliminate parcellation biases, ensemble approaches integrating information across parcellations have been proposed (112,113). Ensembling across predefined parcellations improved the classification of individuals with schizophrenia versus healthy control subjects based on intrinsic functional connectivity and regional activation properties compared with predictions based on a single parcellation (113). Similarly, ensembling across predefined or stochastic parcellations yielded more accurate and robust classification of individuals with autism versus healthy control subjects based on intrinsic functional connectivity than predictions based on a single predefined parcellation (112).

A second alternative involves individual-specific parcellations. These parcellations consider individual differences in spatial arrangement of the brain and are likely to be critical in psychiatric populations with known differences in network

topography (107,109,114,115). These individual-specific network topography patterns have demonstrated improved accuracy for behavioral predictions compared with population-level parcellations (107).

Sample Size and Transfer Learning

Historically, psychiatric neuroimaging studies relied on as few as tens of subjects to characterize interactions between neurobiology and diagnoses/phenotypic presentations (116). A core issue when applying machine learning to neuroimaging is sample size. Generally, machine learning algorithms require sample sizes larger than the dimensionality of the feature set to prevent overfitting (117). Over the past decade, sample sizes have increased to hundreds and even thousands. Limited sample sizes result in decreased reliability of predictions regardless of algorithm, feature type, and phenotype (84,118–120) and produce overly optimistic estimates of prediction accuracy (117,121). These issues are of particular concern in heterogeneous populations, such as individuals with psychiatric illnesses, where small datasets will likely produce models that cannot generalize to independent test samples and thus lack clinical utility. In fact, there appears to be a strong negative relationship between accuracy and sample size for diagnostic and behavioral predictions in Alzheimer's disease, depression, schizophrenia, psychosis, and autism (117,120). However, large clinical samples can be difficult to acquire.

Fortunately, there is evidence to suggest that brain-behavior predictive models developed in the general population can boost predictions of new phenotypes in smaller independent datasets. A meta-matching framework developed to translate predictive models based on neuroimaging features from large samples into nonbrain imaging features in smaller samples has shown promising results when generalizing from aging to young adult populations (122). Connectome-based predictive models of attention have also demonstrated the capacity to generalize across healthy and clinical populations (14,15,123), while models of memory have successfully generalized across psychiatric illnesses (16). Translational applications of these predictive frameworks provide the opportunity to leverage information about brain-behavior relationships in the general population or within a specific clinical population to better understand psychiatric presentation. This can drastically reduce the clinical sample size needed to yield accurate and reliable predictions.

Population-Specific Models

The rise of machine learning applications in neuroscience and psychiatry has begun to expose how (lack of) demographic representation in datasets and algorithmic biases can inadvertently introduce biases in behavioral predictions for specific populations. These discrepancies cannot be entirely attributed to under-representation in datasets or unequal sample sizes and may also be tied to differences in how accurately and reliably neuroimaging and behavioral measures are able to capture phenotypes across populations (124). Evidence suggests that behavioral prediction models trained on datasets dominated by White Americans may fail to generalize to African

American populations and that the underlying brain-behavior associations captured by the models are specifically representative of White Americans (125). Similarly, predictions of cognitive and personality traits can fail to generalize across sexes (88,126–128). Distinct demographic groups (i.e., age groups, sexes, races/ethnicities) often exhibit differences in psychiatric presentations including prevalence, symptom profile, and treatment response. Traditional neuroimaging analyses relied on covariate regression to minimize effects of sex, age, and/or other demographic or data collection-dependent variables (i.e., site). However, common methods to remove covariate information may not sufficiently account for their effects (66,129) and are likely to result in a loss of meaningful variance in the data. Covariate regression, if not performed properly within nested designs, can also lead to information leakage and artifactually inflate model performance and effect sizes. Alternatively, the use of population-specific prediction models can provide insight into how brain-behavior relationships vary across different populations.

While general predictions can capture shared neural correlates of different clinically relevant behaviors, population-specific models can provide complementary information about brain-behavior relationships specific to different groups. Recent work has shown that general prediction models often fail in individuals who do not fit a sample population's stereotypical profile for brain-behavior relationships (130). Demonstrating the value of population-specific predictive modeling, several groups have shown that sex-specific predictions of cognition and personality are more accurate and insightful than sex-independent ones (39,126–128). More specifically, these models reveal the shared and unique correlates of behaviors in men and women. Taken together, these data emphasize the clinical utility of population-specific predictive models. Given the well-established sex differences in prevalence, symptomatology, and clinical trajectory in psychiatric illnesses, sex-specific predictive models will be especially informative in distinct clinical populations. Similarly, precision psychiatry will also likely benefit from models specific to other demographic and clinical variables that can drive differences in disease presentation and treatment-seeking behaviors.

A crucial factor to acknowledge when developing population-specific models is intersectionality. The overlapping and interdependent nature of the way in which distinct demographic and social identities (i.e., age, sex, race, ethnicity, sexuality, physical health, psychiatric diagnoses, socioeconomic status, religion) may intersect is extremely complex, and individuals with psychiatric illnesses may be members of several different marginalized groups (131). Other variables pertaining to data collection (i.e., data quality, site) are also likely to be closely intertwined with participants' intersectional identities. Therefore, when developing population-specific models, researchers must carefully consider how models may generalize across intersectional populations. Models specific to one population may not comparably generalize to other demographic and social groups, but such models may still reveal brain-behavior associations specific to that group that can aid in clinical decision making. Moreover, ensembling predictions across models developed across distinct populations that an

individual belongs to may also provide clinically relevant insights.

CLINICAL UTILITY AND FUTURE DIRECTIONS

An ongoing debate in the field seeks to establish whether predictive models should prioritize accuracy, generalizability, or interpretability. Existing literature suggests that a wide range of modeling choices can be leveraged to yield clinically informative predictions. Models with high accuracy and interpretability but low generalizability may offer important insights into brain-behavior relationships in clinical populations but lack clinical utility. Conversely, models with high accuracy and generalizability but low interpretability may have high clinical utility but are unable to provide clear explanations that can aid clinical decision making. Finally, while models with high interpretability and generalizability but low accuracy are extremely unlikely by definition, they can produce mechanistic insights into the neurobiological features underlying a phenotype. In the future, the field of precision psychiatry must involve interdisciplinary researchers and carefully consider how methodological choices can be used to develop machine learning models that are in line with their broader analytical goals.

Brain-based predictive modeling in psychiatry and neuroscience has led to monumental advances in our understanding of the neurobiological correlates of psychiatric illness. In the next decade, we expect continued growth in this area of research. Two particular areas of focus that are crucial for significant advances include scale development and clinical evaluation. First, the development and implementation of behavioral scales that accurately, reliably, and robustly capture individual differences across distinct clinically relevant behaviors is fundamental. Brain-based behavioral prediction models can be successful only if the behaviors they are optimized to capture are measured correctly. Second, predictive and prescriptive analyses must be clinically evaluated prospectively in appropriately pre-registered randomized controlled clinical trials. A clinical evaluation of these approaches is necessary before we can actualize and implement predictive models in psychiatry.

CONCLUSIONS

Historical advances in our understanding of neurobiological correlates of psychiatric symptoms have paralleled technological advances enabling the visualization and quantification of human brain functions. Future advances can exploit multivariate analytical tools to model accurate and interpretable relationships between brain and behavior at an individual level. Machine learning with neuroimaging data provides a framework that can support innovations in precision neuropsychiatry. However, the development of predictive models to facilitate accurate diagnosis, informative prognosis, and impactful treatment must consider methodological choices that can influence model performance. Here, we reviewed how different factors can affect both the accuracy and the interpretability of these models. An understanding of these effects is crucial for the mindful implementation of choices that are most optimal for a specific research goal. Proper development and deployment of these approaches will yield significant

improvements in our understanding of psychiatric illness and improve the lives of millions of individuals with such illnesses.

ACKNOWLEDGMENTS AND DISCLOSURES

This work was supported by the National Institute of Mental Health (Grant Nos. R01MH120080 and R01MH123245 [to AJH]) and the Kavli Institute for Neuroscience at Yale University (Postdoctoral Fellowship for Academic Diversity [to ED]). This work was also supported by the Singapore National Research Foundation (NRF) Fellowship (Class of 2017) (to BTTY), the NUS Yong Loo Lin School of Medicine (Grant No. NUHSRO/2020/124/TMR/LOA [to BTTY]), the Singapore National Medical Research Council (NMRC) LCG (Grant No. OFLCG19May-0035 [to BTTY]), and the NMRC STaR (Grant No. STaR20nov-0003 [to BTTY]).

Any opinions, findings, and conclusions or recommendations expressed in this material are those of the authors and do not reflect the views of the Singapore NRF or the Singapore NMRC.

The authors report no biomedical financial interests or potential conflicts of interest.

ARTICLE INFORMATION

From the Department of Psychology, Yale University, New Haven, Connecticut (ED, AJH); Kavli Institute for Neuroscience, Yale University, New Haven, Connecticut (ED, AJH); Centre for Sleep & Cognition & Centre for Translational Magnetic Resonance Research, Yong Loo Lin School of Medicine, Singapore, National University of Singapore, Singapore (BTTY); Department of Electrical and Computer Engineering, National University of Singapore, Singapore (BTTY); N.1 Institute for Health & Institute for Digital Medicine, National University of Singapore, Singapore (BTTY); Integrative Sciences and Engineering Programme, National University of Singapore, Singapore (BTTY); Martinos Center for Biomedical Imaging, Massachusetts General Hospital, Charlestown, Massachusetts (BTTY); Department of Psychiatry, Yale University, New Haven, Connecticut (AJH); and Wu Tsai Institute, Yale University, New Haven, Connecticut (AJH).

Address correspondence to Elvisha Dhamala, Ph.D., at elvisha.dhamala@yale.edu, or Avram J. Holmes, Ph.D., at avram.holmes@yale.edu.

Received May 13, 2022; revised Sep 7, 2022; accepted Sep 23, 2022.

REFERENCES

- National Research Council (2011): Toward Precision Medicine: Building a Knowledge Network for Biomedical Research and a New Taxonomy of Disease. Washington, DC: The National Academies Press.
- Dugger SA, Platt A, Goldstein DB (2018): Drug development in the era of precision medicine. *Nat Rev Drug Discov* 17:183–196.
- Vieta E (2015): Personalised medicine applied to mental health: Precision psychiatry. *Rev Psychiatr Salud Ment* 8:117–118.
- Hyman SE (2010): The diagnosis of mental disorders: The problem of reification. *Annu Rev Clin Psychol* 6:155–179.
- Yip SW, Kiluk B, Scheinost D (2020): Toward addiction prediction: An overview of cross-validated predictive modeling findings and considerations for future neuroimaging research. *Biol Psychiatry Cogn Neurosci Neuroimaging* 5:748–758.
- Scheinost D, Noble S, Horien C, Greene AS, Lake EM, Salehi M, *et al.* (2019): Ten simple rules for predictive modeling of individual differences in neuroimaging. *NeuroImage* 193:35–45.
- Rosenberg MD, Casey BJ, Holmes AJ (2018): Prediction complements explanation in understanding the developing brain. *Nat Commun* 9:589.
- Varoquaux G, Raamana PR, Engemann DA, Hoyos-Idrobo A, Schwartz Y, Thirion B (2017): Assessing and tuning brain decoders: Cross-validation, caveats, and guidelines. *NeuroImage* 145:166–179.
- Woo CW, Chang LJ, Lindquist MA, Wager TD (2017): Building better biomarkers: Brain models in translational neuroimaging. *Nat Neurosci* 20:365–377.
- Arbabshirani MR, Plis S, Sui J, Calhoun VD (2017): Single subject prediction of brain disorders in neuroimaging: Promises and pitfalls. *NeuroImage* 145:137–165.
- Khosla M, Jamison K, Ngo GH, Kuceyeski A, Sabuncu MR (2019): Machine learning in resting-state fMRI analysis. *Magn Reson Imaging* 64:101–121.
- Shen X, Finn ES, Scheinost D, Rosenberg MD, Chun MM, Papademetris X, Constable RT (2017): Using connectome-based predictive modeling to predict individual behavior from brain connectivity. *Nat Protoc* 12:506–518.
- Kawahara J, Brown CJ, Miller SP, Booth BG, Chau V, Grunau RE, *et al.* (2017): BrainNetCNN: Convolutional neural networks for brain networks; towards predicting neurodevelopment. *Neuroimage* 146:1038–1049.
- Yoo K, Rosenberg MD, Hsu WT, Zhang S, Li C-SR, Scheinost D, *et al.* (2018): Connectome-based predictive modeling of attention: Comparing different functional connectivity features and prediction methods across datasets. *Neuroimage* 167:11–22.
- Rosenberg MD, Hsu WT, Scheinost D, Todd Constable R, Chun MM (2018): Connectome-based models predict separable components of attention in novel individuals. *J Cogn Neurosci* 30:160–173.
- Barron DS, Gao S, Dadashkarimi J, Greene AS, Spann MN, Noble S, *et al.* (2021): Transdiagnostic, connectome-based prediction of memory constructs across psychiatric disorders. *Cereb Cortex* 31:2523–2533.
- Wang Z, Goerlich KS, Ai H, Aleman A, Luo YJ, Xu P (2021): Connectome-based predictive modeling of individual anxiety. *Cereb Cortex* 31:3006–3020.
- Ju Y, Horien C, Chen W, Guo W, Lu X, Sun J, *et al.* (2020): Connectome-based models can predict early symptom improvement in major depressive disorder. *J Affect Disord* 273:442–452.
- Yip SW, Scheinost D, Potenza MN, Carroll KM (2019): Connectome-based prediction of cocaine abstinence. *Am J Psychiatry* 176:156–164.
- Lichenstein SD, Scheinost D, Potenza MN, Carroll KM, Yip SW (2021): Dissociable neural substrates of opioid and cocaine use identified via connectome-based modelling. *Mol Psychiatry* 26:4383–4393.
- He T, Kong R, Holmes AJ, Nguyen M, Sabuncu MR, Eickhoff SB, *et al.* (2020): Deep neural networks and kernel regression achieve comparable accuracies for functional connectivity prediction of behavior and demographics. *NeuroImage* 206:116276.
- Pervais U, Vidaurre D, Woolrich MW, Smith SM (2020): Optimising network modelling methods for fMRI. *Neuroimage* 211:116604.
- Abrol A, Fu Z, Salman M, Silva R, Du Y, Plis S, Calhoun V (2021): Deep learning encodes robust discriminative neuroimaging representations to outperform standard machine learning. *Nat Commun* 12:353.
- Mellema CJ, Nguyen KP, Treacher A, Montillo A (2022): Reproducible neuroimaging features for diagnosis of autism spectrum disorder with machine learning. *Sci Rep* 12:3057.
- Traut N, Heuer K, Lemaître G, Beggato A, Germanaud D, Elmaleh M, *et al.* (2022): Insights from an autism imaging biomarker challenge: Promises and threats to biomarker discovery. *NeuroImage* 255:119171.
- Dadi K, Rahim M, Abraham A, Chyzyk D, Milham M, Thirion B, *et al.* (2019): Benchmarking functional connectome-based predictive models for resting-state fMRI. *NeuroImage* 192:115–134.
- Dhamala E, Jamison KW, Jaywant A, Dennis S, Kuceyeski A (2021): Distinct functional and structural connections predict crystallised and fluid cognition in healthy adults. *Hum Brain Mapp* 42:3102–3118.
- Mansour LSL, Tian Y, Yeo BTT, Cropley V, Zalesky A (2021): High-resolution connectomic fingerprints: Mapping neural identity and behavior. *NeuroImage* 229:117695.
- Ooi LQR, Chen J, Zhang S, Kong R, Li J, Dhamala E, *et al.* (2022): Comparison of individualized behavioral predictions across anatomical, diffusion and functional connectivity MRI. *Neuroimage* 263:119636.
- Mellem MS, Liu Y, Gonzalez H, Kollada M, Martin WJ, Ahammad P (2020): Machine learning models identify multimodal measurements highly predictive of transdiagnostic symptom severity for mood, anhedonia, and anxiety. *Biol Psychiatry Cogn Neurosci Neuroimaging* 5:56–67.

31. Seguin C, Tian Y, Zalesky A (2020): Network communication models improve the behavioral and functional predictive utility of the human structural connectome. *Netw Neurosci* 4:980–1006.
32. Sabuncu MR, Konukoglu E, Alzheimer's Disease Neuroimaging Initiative (2015): Clinical prediction from structural brain MRI scans: A large-scale empirical study. *Neuroinformatics* 13:31–46.
33. Talpalari A, Bhagwat N, Devenyi GA, Lepage M, Chakravarty MM (2019): Identifying schizophrenia subgroups using clustering and supervised learning. *Schizophr Res* 214:51–59.
34. Koutsouleris N, Wobrock T, Guse B, Langguth B, Landgrebe M, Eichhammer P, *et al.* (2018): Predicting response to repetitive transcranial magnetic stimulation in patients with schizophrenia using structural magnetic resonance imaging: A multisite machine learning analysis. *Schizophr Bull* 44:1021–1034.
35. Johnston BA, Steele JD, Tolomeo S, Christmas D, Matthews K (2015): Structural MRI-based predictions in patients with treatment-refractory depression (TRD). *PLoS One* 10:e0132958.
36. Whitfield-Gabrieli S, Ghosh SS, Nieto-Castanon A, Saygin Z, Doehrmann O, Chai XJ, *et al.* (2016): Brain connectomics predict response to treatment in social anxiety disorder. *Mol Psychiatry* 21:680–685.
37. Johnson KA, Duffley G, Anderson DN, Ostrem JL, Welter ML, Baldermann JC, *et al.* (2020): Structural connectivity predicts clinical outcomes of deep brain stimulation for Tourette syndrome. *Brain* 143:2607–2623.
38. Shao J, Myers N, Yang Q, Feng J, Plant C, Böhm C, *et al.* (2012): Prediction of Alzheimer's disease using individual structural connectivity networks. *Neurobiol Aging* 33:2756–2765.
39. Greene AS, Gao S, Scheinost D, Constable RT (2018): Task-induced brain state manipulation improves prediction of individual traits. *Nat Commun* 9:2807.
40. Chen J, Tam A, Kebets V, Orban C, Ooi LQR, Asplund CL, *et al.* (2022): Shared and unique brain network features predict cognitive, personality, and mental health scores in the ABCD study. *Nat Commun* 13:2217.
41. Sarpal DK, Argyelan M, Robinson DG, Szeszko PR, Karlsgodt KH, John M, *et al.* (2016): Baseline striatal functional connectivity as a predictor of response to antipsychotic drug treatment. *Am J Psychiatry* 173:69–77.
42. Du Y, Fu Z, Sui J, Gao S, Xing Y, Lin D, *et al.* (2020): NeuroMark: An automated and adaptive ICA based pipeline to identify reproducible fMRI markers of brain disorders. *NeuroImage Clin* 28:102375.
43. Pantazatos SP, Talati A, Schneier FR, Hirsch J (2014): Reduced anterior temporal and hippocampal functional connectivity during face processing discriminates individuals with social anxiety disorder from healthy controls and panic disorder, and increases following treatment. *Neuropsychopharmacology* 39:425–434.
44. Calhoun VD, Maciejewski PK, Pearlson GD, Kiehl KA (2008): Temporal lobe and "default" hemodynamic brain modes discriminate between schizophrenia and bipolar disorder. *Hum Brain Mapp* 29:1265–1275.
45. Cha J, Speaker S, Hu B, Altinay M, Koirala P, Karne H, *et al.* (2021): Neuroimaging correlates of emotional response-inhibition discriminate between young depressed adults with and without sub-threshold bipolar symptoms (Emotional Response-inhibition in Young Depressed Adults). *J Affect Disord* 281:303–311.
46. Doehrmann O, Ghosh SS, Polli FE, Reynolds GO, Horn F, Keshavan A, *et al.* (2013): Predicting treatment response in social anxiety disorder from functional magnetic resonance imaging. *JAMA Psychiatry* 70:87–97.
47. Hahn T, Kircher T, Straube B, Wittchen HU, Konrad C, Ströhle A, *et al.* (2015): Predicting treatment response to cognitive behavioral therapy in panic disorder with agoraphobia by integrating local neural information. *JAMA Psychiatry* 72:68–74.
48. Greene DJ, Black KJ, Schlaggar BL (2016): Considerations for MRI study design and implementation in pediatric and clinical populations. *Dev Cogn Neurosci* 18:101–112.
49. Power JD, Barnes KA, Snyder AZ, Schlaggar BL, Petersen SE (2012): Spurious but systematic correlations in functional connectivity MRI networks arise from subject motion. *Neuroimage* 59:2142–2154.
50. Power JD, Mitra A, Laumann TO, Snyder AZ, Schlaggar BL, Petersen SE (2014): Methods to detect, characterize, and remove motion artifact in resting state fMRI. *Neuroimage* 84:320–341.
51. Power JD, Schlaggar BL, Petersen SE (2015): Recent progress and outstanding issues in motion correction in resting state fMRI. *Neuroimage* 105:536–551.
52. Tavor I, Parker Jones OP, Mars RB, Smith SM, Behrens TE, Jbabdi S (2016): Task-free MRI predicts individual differences in brain activity during task performance. *Science* 352:216–220.
53. Gruskin DC, Patel GH (2022): Brain connectivity at rest predicts individual differences in normative activity during movie watching. *NeuroImage* 253:119100.
54. Gal S, Tik N, Bernstein-Eliav M, Tavor I (2022): Predicting individual traits from unperformed tasks. *NeuroImage* 249:118920.
55. Botvinik-Nezer R, Holzmeister F, Camerer CF, Dreber A, Huber J, Johannesson M, *et al.* (2020): Variability in the analysis of a single neuroimaging dataset by many teams. *Nature* 582:84–88.
56. Bhagwat N, Barry A, Dickie EW, Brown ST, Devenyi GA, Hatano K, *et al.* (2021): Understanding the impact of preprocessing pipelines on neuroimaging cortical surface analyses. *GigaScience* 10:giaa155.
57. Bowring A, Nichols TE, Maumet C (2022): Isolating the sources of pipeline-variability in group-level task-fMRI results. *Hum Brain Mapp* 43:1112–1128.
58. Bowring A, Maumet C, Nichols TE (2019): Exploring the impact of analysis software on task fMRI results. *Hum Brain Mapp* 40:3362–3384.
59. Thomas RM, Bruin W, Zhutovsky P, van Wingen G (2020): Dealing with missing data, small sample sizes, and heterogeneity in machine learning studies of brain disorders. *Machine Learning*. London: Elsevier: 249–266.
60. Murphy K, Fox MD (2017): Towards a consensus regarding global signal regression for resting state functional connectivity MRI. *Neuroimage* 154:169–173.
61. Liu TT, Nalci A, Falahpour M (2017): The global signal in fMRI: Nuisance or Information? *Neuroimage* 150:213–229.
62. Li JW, Kong R, Liégeois R, Orban C, Tan YR, Sun NB, *et al.* (2019): Global signal regression strengthens association between resting-state functional connectivity and behavior. *Neuroimage* 196:126–141.
63. Descoteaux M, Deriche R, Knösche TR, Anwander A (2009): Deterministic and probabilistic tractography based on complex fibre orientation distributions. *IEEE Trans Med Imaging* 28:269–286.
64. Sarwar T, Ramamohanarao K, Zalesky A (2019): Mapping connectomes with diffusion MRI: Deterministic or probabilistic tractography? *Magn Reson Med* 81:1368–1384.
65. Buckner RL, Head D, Parker J, Fotenos AF, Marcus D, Morris JC, Snyder AZ (2004): A unified approach for morphometric and functional data analysis in young, old, and demented adults using automated atlas-based head size normalization: Reliability and validation against manual measurement of total intracranial volume. *Neuroimage* 23:724–738.
66. Dhamala E, Ooi LQR, Chen J, Kong R, Anderson KM, Chin R, *et al.* (2022): Proportional intracranial volume correction differentially biases behavioral predictions across neuroanatomical features, sexes, and development. *NeuroImage* 260:119485.
67. Marquand AF, Rezek I, Buitelaar J, Beckmann CF (2016): Understanding heterogeneity in clinical cohorts using normative models: Beyond case-control studies. *Biol Psychiatry* 80:552–561.
68. Parkes L, Moore TM, Calkins ME, Cook PA, Cieslak M, Roalf DR, *et al.* (2021): Transdiagnostic dimensions of psychopathology explain individuals' unique deviations from normative neurodevelopment in brain structure. *Transl Psychiatry* 11:232.
69. Yang GJ, Murray JD, Repovs G, Cole MW, Savic A, Glasser MF, *et al.* (2014): Altered global brain signal in schizophrenia. *Proc Natl Acad Sci U S A* 111:7438–7443.

70. Klein M, Walters RK, Demontis D, Stein JL, Hibar DP, Adams HH, *et al.* (2019): Genetic markers of ADHD-related variations in intracranial volume. *Am J Psychiatry* 176:228–238.
71. Ohi K, Shimada T, Kataoka Y, Yasuyama T, Kawasaki Y, Shioiri T, Thompson PM (2020): Genetic correlations between subcortical brain volumes and psychiatric disorders. *Br J Psychiatry* 216:280–283.
72. Churchwell JC, Lopez-Larson M, Yurgelun-Todd DA (2010): Altered frontal cortical volume and decision making in adolescent cannabis users. *Front Psychol* 1:225.
73. Wang X, Xie H, Chen T, Cotton AS, Salminen LE, Logue MW, *et al.* (2021): Cortical volume abnormalities in posttraumatic stress disorder: An ENIGMA-psychiatric genomics consortium PTSD workgroup mega-analysis. *Mol Psychiatry* 26:4331–4343.
74. Kuperberg GR, Broome MR, McGuire PK, David AS, Eddy M, Ozawa F, *et al.* (2003): Regionally localized thinning of the cerebral cortex in schizophrenia. *Arch Gen Psychiatry* 60:878–888.
75. Makris N, Goldstein JM, Kennedy D, Hodge SM, Caviness VS, Faraone SV, *et al.* (2006): Decreased volume of left and total anterior insular lobule in schizophrenia. *Schizophr Res* 83:155–171.
76. Shorter E, van Praag HM (2010): Disease versus dimension in diagnosis. *Can J Psychiatry* 55:59–64.
77. Waszczuk MA, Zimmerman M, Ruggero C, Li K, MacNamara A, Weinberg A, *et al.* (2017): What do clinicians treat: Diagnoses or symptoms? The incremental validity of a symptom-based, dimensional characterization of emotional disorders in predicting medication prescription patterns. *Compr Psychiatry* 79:80–88.
78. Clark LA, Cuthbert B, Lewis-Fernández R, Narrow WE, Reed GM (2017): Three approaches to understanding and classifying mental disorder: ICD-11, DSM-5, and the National Institute of Mental Health's Research Domain Criteria (RDoC). *Psychol Sci Public Interest* 18:72–145.
79. van Heugten-van der Kloet D, van Heugten T (2015): The classification of psychiatric disorders according to DSM-5 deserves an internationally standardized psychological test battery on symptom level. *Front Psychol* 6:1108.
80. Freedman R, Lewis DA, Michels R, Pine DS, Schultz SK, Tamminga CA, *et al.* (2013): The initial field trials of DSM-5: New blooms and old thorns. *Am J Psychiatry* 170:1–5.
81. Galatzer-Levy IR, Bryant RA (2013): 636,120 Ways to have post-traumatic stress disorder. *Perspect Psychol Sci* 8:651–662.
82. Saggat M, Uddin LQ (2019): Pushing the boundaries of psychiatric neuroimaging to ground diagnosis in biology. *eNeuro* 6.
83. Alexander LM, Salum GA, Swanson JM, Milham MP (2017): Balancing strengths and weaknesses in dimensional psychiatry. *bioRxiv*. <https://doi.org/10.1101/207019>.
84. Marek S, Tervo-Clemmens B, Calabro FJ, Montez DF, Kay BP, Hatoum AS, *et al.* (2022): Reproducible brain-wide association studies require thousands of individuals [published correction appears in *Nature* 2022; 605(7911):E11]. *Nature* 603(7902):654–660.
85. Carlozzi NE, Tulsy DS, Wolf TJ, Goodnight S, Heaton RK, Casaletto KB, *et al.* (2017): Construct validity of the NIH Toolbox cognition battery in individuals with stroke. *Rehabil Psychol* 62:443–454.
86. Heaton RK, Akshoomoff N, Tulsy D, Mungas D, Weintraub S, Dikmen S, *et al.* (2014): Reliability and validity of composite scores from the NIH Toolbox Cognition Battery in adults. *J Int Neuropsychol Soc* 20:588–598.
87. Weintraub S, Dikmen SS, Heaton RK, Tulsy DS, Zelazo PD, Slotkin J, *et al.* (2014): The cognition battery of the NIH toolbox for assessment of neurological and behavioral function: Validation in an adult sample. *J Int Neuropsychol Soc* 20:567–578.
88. Dhamala E, Jamison KW, Jaywant A, Kuceyeski A (2022): Shared functional connections within and between cortical networks predict cognitive abilities in adult males and females. *Hum Brain Mapp* 43:1087–1102.
89. Goldstein-Piekarski AN, Ball TM, Samara Z, Staveland BR, Keller AS, Fleming SL, *et al.* (2022): Mapping neural circuit biotypes to symptoms and behavioral dimensions of depression and anxiety. *Biol Psychiatry* 91:561–571.
90. Eickhoff SB, Yeo BTT, Genov S (2018): Imaging-based parcellations of the human brain. *Nat Rev Neurosci* 19:672–686.
91. Uddin LQ, Betzel RF, Cohen JR, Damoiseaux JS, De Brigard F, Eickhoff SB, *et al.* (2022): Controversies and current progress on large-scale brain network nomenclature from OHBM WHATNET: Workgroup for HARmonized Taxonomy of NETWORKs. *OSF Preprints*. <https://doi.org/10.31219/osf.io/25za6>.
92. Amunts K, Mohlberg H, Bludau S, Zilles K (2020): Julich-Brain: A 3D probabilistic atlas of the human brain's cytoarchitecture. *Science* 369:988–992.
93. Eickhoff SB, Rottschy C, Kujovic M, Palomero-Gallagher N, Zilles K (2008): Organizational principles of human visual cortex revealed by receptor mapping. *Cereb Cortex* 18:2637–2645.
94. Eickhoff SB, Stephan KE, Mohlberg H, Grefkes C, Fink GR, Amunts K, Zilles K (2005): A new SPM toolbox for combining probabilistic cytoarchitectonic maps and functional imaging data. *Neuroimage* 25:1325–1335.
95. Destrieux C, Fischl B, Dale A, Hagren E (2010): Automatic parcellation of human cortical gyri and sulci using standard anatomical nomenclature. *Neuroimage* 53:1–15.
96. Talairach J (1988): Co-planar Stereotaxic Atlas of the Human Brain-3-Dimensional Proportional System: An Approach to Cerebral Imaging. Stuttgart: Thieme.
97. Desikan RS, Ségonne F, Fischl B, Quinn BT, Dickerson BC, Blacker D, *et al.* (2006): An automated labeling system for subdividing the human cerebral cortex on MRI scans into gyral based regions of interest. *Neuroimage* 31:968–980.
98. Schaefer A, Kong R, Gordon EM, Laumann TO, Zuo XN, Holmes AJ, *et al.* (2018): Local-global parcellation of the human cerebral cortex from intrinsic functional connectivity MRI. *Cereb Cortex* 28:3095–3114.
99. Yeo BT, Krienen FM, Sepulcre J, Sabuncu MR, Lashkari D, Hollinshead M, *et al.* (2011): The organization of the human cerebral cortex estimated by intrinsic functional connectivity. *J Neurophysiol* 106:1125–1165.
100. Craddock RC, James GA, Holtzheimer PE 3rd, Hu XP, Mayberg HS (2012): A whole brain fMRI atlas generated via spatially constrained spectral clustering. *Hum Brain Mapp* 33:1914–1928.
101. Power JD, Cohen AL, Nelson SM, Wig GS, Barnes KA, Church JA, *et al.* (2011): Functional network organization of the human brain. *Neuron* 72:665–678.
102. Dosenbach NU, Nardos B, Cohen AL, Fair DA, Power JD, Church JA, *et al.* (2010): Prediction of individual brain maturity using fMRI. *Science* 329:1358–1361.
103. Bellec P, Rosa-Neto P, Lyttelton OC, Benali H, Evans AC (2010): Multi-level bootstrap analysis of stable clusters in resting-state fMRI. *Neuroimage* 51:1126–1139.
104. Smith SM, Fox PT, Miller KL, Glahn DC, Fox PM, Mackay CE, *et al.* (2009): Correspondence of the brain's functional architecture during activation and rest. *Proc Natl Acad Sci U S A* 106:13040–13045.
105. Wu L, Calhoun VD, Jung RE, Caprihan A (2015): Connectivity-based whole brain dual parcellation by group ICA reveals tract structures and decreased connectivity in schizophrenia. *Hum Brain Mapp* 36:4681–4701.
106. Ji Y, Hervé P-Y, Aickelin U, Pitiot A (2009): Parcellation of fMRI datasets with ICA and PLS—a data driven approach. *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2009*. Berlin: Springer: 984–991.
107. Kong R, Li J, Orban C, Sabuncu MR, Liu H, Schaefer A, *et al.* (2019): Spatial topography of individual-specific cortical networks predicts human cognition, personality, and emotion. *Cereb Cortex* 29:2533–2551.
108. Thirion B, Varoquaux G, Dohmatob E, Poline JB (2014): Which fMRI clustering gives good brain parcellations? *Front Neurosci* 8:167.
109. Kong R, Yang Q, Gordon E, Xue A, Yan X, Orban C, *et al.* (2021): Individual-specific areal-level parcellations improve functional connectivity prediction of behavior. *Cereb Cortex* 31:4477–4500.

110. Dhamala E, Jamison KW, Sabuncu MR, Kuceyeski A (2020): Sex classification using long-range temporal dependence of resting-state functional MRI time series. *Hum Brain Mapp* 41:3567–3579.
111. Zeighami Y, Evans AC (2021): Association vs. prediction: The impact of cortical surface smoothing and parcellation on brain age. *Front Big Data* 4:637724.
112. Khosla M, Jamison K, Kuceyeski A, Sabuncu MR (2019): Ensemble learning with 3D convolutional neural networks for functional connectome-based prediction. *Neuroimage* 199:651–662.
113. Kalmady SV, Greiner R, Agrawal R, Shivakumar V, Narayanaswamy JC, Brown MRG, *et al.* (2019): Towards artificial intelligence in mental health by improving schizophrenia prediction with multiple brain parcellation ensemble-learning. *npj Schizophr* 5:2.
114. Gordon EM, Laumann TO, Gilmore AW, Newbold DJ, Greene DJ, Berg JJ, *et al.* (2017): Precision functional mapping of individual human brains. *Neuron* 95:791–807.e7.
115. Gratton C, Laumann TO, Nielsen AN, Greene DJ, Gordon EM, Gilmore AW, *et al.* (2018): Functional brain networks are dominated by stable group and individual factors, not cognitive or daily variation. *Neuron* 98:439–452.e5.
116. Szucs D, Ioannidis JP (2020): Sample size evolution in neuroimaging research: An evaluation of highly-cited studies (1990–2012) and of latest practices (2017–2018) in high-impact journals. *Neuroimage* 221:117164.
117. Janssen RJ, Mourão-Miranda J, Schnack HG (2018): Making individual prognoses in psychiatry using neuroimaging and machine learning. *Biol Psychiatry Cogn Neurosci Neuroimaging* 3:798–808.
118. Cui Z, Gong G (2018): The effect of machine learning regression algorithms and sample size on individualized behavioral prediction with functional connectivity features. *Neuroimage* 178:622–637.
119. Jollans L, Boyle R, Artiges E, Banaschewski T, Desrivières S, Grigis A, *et al.* (2019): Quantifying performance of machine learning methods for neuroimaging data. *Neuroimage* 199:351–365.
120. Varoquaux G (2018): Cross-validation failure: Small sample sizes lead to large error bars. *Neuroimage* 180:68–77.
121. Santos MS, Soares JP, Abreu PH, Araujo H, Santos J (2018): Cross-validation for imbalanced datasets: Avoiding overoptimistic and overfitting approaches. *IEEE Comput Intell M* 13:59–76.
122. He T, An L, Chen P, Chen J, Feng J, Bzdok D, *et al.* (2022): Meta-matching as a simple framework to translate phenotypic predictive models from big to small data. *Nat Neurosci* 25:795–804.
123. Rosenberg MD, Finn ES, Scheinost D, Papademetris X, Shen X, Constable RT, Chun MM (2016): A neuromarker of sustained attention from whole-brain functional connectivity. *Nat Neurosci* 19:165–171.
124. Ricard JA, Parker TC, Dhamala E, Kwasa J, Allsop AS, Holmes AJ (2023): Confronting racially exclusionary practices in the acquisition and analyses of neuroimaging data. *Nat Neurosci* 26:4–11.
125. Li J, Bzdok D, Chen J, Tam A, Ooi LQR, Holmes AJ, *et al.* (2022): Cross-ethnicity/race generalization failure of behavioral prediction from resting-state functional connectivity. *Sci Adv* 8:eabj1812.
126. Jiang R, Calhoun VD, Cui Y, Qi S, Zhuo C, Li J, *et al.* (2020): Multimodal data revealed different neurobiological correlates of intelligence between males and females. *Brain Imaging Behav* 14:1979–1993.
127. Jiang R, Calhoun VD, Fan L, Zuo N, Jung R, Qi S, *et al.* (2020): Gender differences in connectome-based predictions of individualized intelligence quotient and sub-domain scores. *Cereb Cortex* 30:888–900.
128. Nostro AD, Müller VI, Varikuti DP, Pläschke RN, Hoffstaedter F, Langner R, *et al.* (2018): Predicting personality from network-based resting-state functional connectivity. *Brain Struct Funct* 223:2699–2719.
129. Sanchis-Segura C, Ibañez-Gual MV, Aguirre N, Cruz-óñez AJ, Forn C (2020): Effects of different intracranial volume correction methods on univariate sex differences in grey matter volume and multivariate sex prediction. *Sci Rep* 10:12953.
130. Greene AS, Shen X, Noble S, Horien C, Hahn CA, Arora J, *et al.* (2022): Brain-phenotype models fail for individuals who defy sample stereotypes. *Nature* 609:109–118.
131. Oexle N, Corrigan PW (2018): Understanding mental illness stigma toward persons with multiple stigmatized conditions: Implications of intersectionality theory. *Psychiatr Serv* 69:587–589.
132. Tang S, Sun N, Floris DL, Zhang X, Di Martino A, Yeo BTT (2020): Reconciling dimensional and categorical models of autism heterogeneity: A brain connectomics and behavioral study. *Biol Psychiatry* 87:1071–1082.
133. Hong SJ, Valk SL, Di Martino A, Milham MP, Bernhardt BC (2018): Multidimensional neuroanatomical subtyping of autism spectrum disorder. *Cereb Cortex* 28:3578–3588.
134. Drysdale AT, Grosenick L, Downar J, Dunlop K, Mansouri F, Meng Y, *et al.* (2017): Resting-state connectivity biomarkers define neurophysiological subtypes of depression. *Nat Med* 23:28–38.
135. Wang Y, Tang S, Zhang L, Bu X, Lu L, Li H, *et al.* (2021): Data-driven clustering differentiates subtypes of major depressive disorder with distinct brain connectivity and symptom features. *Br J Psychiatry* 219:606–613.
136. Clementz BA, Sweeney JA, Hamm JP, Ivleva EI, Ethridge LE, Pearlson GD, *et al.* (2016): Identification of distinct psychosis biotypes using brain-based biomarkers. *Am J Psychiatry* 173:373–384.
137. Chen J, Patil KR, Weis S, Sim K, Nickl-Jockschat T, Zhou J, *et al.* (2020): Neurobiological divergence of the positive and negative schizophrenia subtypes identified on a new factor structure of psychopathology using non-negative factorization: An international machine learning study. *Biol Psychiatry* 87:282–293.
138. Yan W, Zhao M, Fu Z, Pearlson GD, Sui J, Calhoun VD (2022): Mapping relationships among schizophrenia, bipolar and schizoaffective disorders: A deep classification and clustering framework using fMRI time series. *Schizophr Res* 245:141–150.
139. Chen J, Müller VI, Dukart J, Hoffstaedter F, Baker JT, Holmes AJ, *et al.* (2021): Intrinsic connectivity patterns of task-defined brain networks allow individual prediction of cognitive symptom dimension of schizophrenia and are linked to molecular architecture. *Biol Psychiatry* 89:308–319.
140. Wang D, Li M, Wang M, Schoeppe F, Ren J, Chen H, *et al.* (2020): Individual-specific functional connectivity markers track dimensional and categorical features of psychotic illness. *Mol Psychiatry* 25:2119–2129.
141. Kebets V, Holmes AJ, Orban C, Tang S, Li J, Sun N, *et al.* (2019): Somatosensory-motor dysconnectivity spans multiple transdiagnostic dimensions of psychopathology. *Biol Psychiatry* 86:779–791.
142. Xia CH, Ma Z, Ciric R, Gu S, Betzel RF, Kaczkurkin AN, *et al.* (2018): Linked dimensions of psychopathology and connectivity in functional brain networks. *Nat Commun* 9:3003.
143. Kaczkurkin AN, Sotiras A, Baller EB, Barzilay R, Calkins ME, Chand GB, *et al.* (2020): Neurostructural heterogeneity in youths with internalizing symptoms. *Biol Psychiatry* 87:473–482.
144. Zhao X, Rangaprakash D, Yuan B, Denney TS Jr, Katz JS, Dretsches MN, Deshpande G (2018): Investigating the correspondence of clinical diagnostic grouping with underlying neurobiological and phenotypic clusters using unsupervised machine learning. *Front Appl Math Stat* 4.
145. Nielsen AN, Barch DM, Petersen SE, Schlaggar BL, Greene DJ (2020): Machine learning with neuroimaging: Evaluating its applications in psychiatry. *Biol Psychiatry Cogn Neurosci Neuroimaging* 5:791–798.
146. Sui J, Jiang R, Bustillo J, Calhoun V (2020): Neuroimaging-based individualized prediction of cognition and behavior for mental disorders and health: Methods and promises. *Biol Psychiatry* 88:818–828.